

Rolando J. Acosta Nuñez, Manager Biostatistics

EDUCATION

05/2022 **PhD, Biostatistics**
Harvard University, Cambridge, MA

05/2020 **MA, Biostatistics**
Harvard University, Cambridge, MA

05/2017 **BS, Computational Mathematics**
University of Puerto Rico, Humacao, PR

EXPERIENCE

01/2024 – Present **Regeneron Pharmaceuticals, Inc., Tarrytown, NY**
Manager Biostatistics, Biostatistics & Data Management
I am an experienced Biostatistician with a proven track record in designing and analyzing clinical studies, leveraging advanced statistical methodologies and artificial intelligence (AI) to drive impactful decision-making in pharmaceutical development. I currently serve at the intersection of biostatistics, clinical trials, and AI as an AI development and strategy lead within the Quantitative Innovation & Statistical Strategy (QISS) group, where I drive the integration of AI and automation into biostatistics workflows. A cross-functional leader, I bridge deep statistical expertise with strategic communication, with a focus on accelerating drug development timelines and advancing therapeutic breakthroughs through innovative methodology and AI adoption. Previously, I split my time between the Biostatistical Engineering team and Clinical Hematology, developing novel digital endpoints through the assessment of wearable devices and overseeing the design, analysis, and execution of clinical trials in Hematology Oncology.

07/2022 – 12/2023 **Regeneron Pharmaceuticals, Inc., Tarrytown, NY**
Principal Biostatistics, Biostatistics & Data Management

- Lead the design and analysis of pilot studies in healthy populations, and first-in-human studies in populations of interest, to assess the operating characteristics of novel wearable devices, driving the development of innovative digital endpoints.
- Advanced pharmaceutical innovations through cutting-edge statistical, machine learning, and artificial intelligence methodologies.
- Collaborated extensively with cross-functional teams, including medical, clinical, operations, and regulatory departments to drive project outcomes.

12/2023 – 03/2025 **Digital Medicine Society, Boston, MA**
Statistical Advisory Committee (SAC) Member

- Served as a SAC member for the [Validating Novel Digital Clinical Measures](#) project.

06/2020, 06/2021 and
06/2022 **Interamerican University, San Germán, PR**
Summer Workshop Instructor

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- Design and implemented data visualization in R and data wrangling in R courses for the Interamerican university data science summer institute.
- [Course webpage](#)
- 09/2017 – 05/2022 **Harvard University, Cambridge, MA**
Ph.D. Student, Biostatistics
 - Development of statistical methods to monitor human mobility and mortality in the aftermath of natural hazards.
- 06/2021 – 08/2021 **Alphabet Google, Remote (Covid-19 Pandemic), MA**
Ph.D. Data Science Intern, Google Payments
 - Optimal Search of Heterogeneous Treatment Effects (OSHTe)
- 09/2019 – 12/2019 and 09/2021 – 12/2021 **Harvard University, Cambridge, MA**
Teaching Assistant, Biostatistics
 - BST 260: Introduction to Data Science
 - [Prof. Heather Mattie](#)
- 01/2020 – 05/2020 and 01/2021 – 05/2021 **Harvard University, Cambridge, MA**
Teaching Assistant, Biostatistics
 - BST 226: Applied Longitudinal Analysis
 - [Prof. Garret Fitzmaurice](#)
- 06/2020 – 10/2020 **Covid-19 Mobility Network, Remote (Covid-19 Pandemic), MA**
Affiliated Researcher
 - Analyzed mobility patterns for the City of Miami to ascertain the effectiveness of social distancing policies.
 - [Group webpage](#)
- 09/2019 – 05/2020 **Harvard University, Cambridge, MA**
Biostatistics Ambassador, Center for Climate, Health, And the Global Environment (C-CHANGE)
 - Biostatistics student ambassador to C-CHANGE
 - [Group webpage](#)
- 09/2019 – 05/2020 **Harvard University, Cambridge, MA**
Statistics Consultant, Biostatistics Student Consulting Center (BSCC)
 - Biostatistics student consultant for the BSCC
 - [Group webpage](#)
- 09/2018 – 12/2018 **Harvard University, Cambridge, MA**
Teaching Assistant, Biostatistics
 - BST 201: Introduction to Statistical Methods
 - [Prof. Paul Catalano](#)
- 03/2018 **University of Puerto Rico, San Juan, PR**
Teaching Assistant, Biostatistics
 - Big Data in Biology: From Genes to the Biosphere
 - [Prof. Rafael Irizarry](#)
 - [Course webpage](#)

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- 01/2015 – 05/2017 **University of Puerto Rico, Humacao, PR**
Undergraduate Research Scientist, Mathematics
 - Explore models for dengue control in Puerto Rico using genetically modified mosquitoes.
- 06/2016 – 08/2016 **Harvard University, Cambridge, MA**
Summer Undergraduate Research Intern, Biomedical Informatics
 - A geo-temporal database for the external exposome.
- 06/2015 – 05/2015 **Harvard University, Cambridge, MA**
Summer Undergraduate Research Intern, Biostatistics
 - Automated phenotyping of patient EMR data: Feature extraction and selection.
- 06/2014 – 05/2014 **University of Iowa, Iowa City, IA**
Summer Undergraduate Research Intern, Biostatistics
 - Assessing the effect of practical considerations when using the CRM in dose finding studies.

PUBLICATIONS

- Robertson EE, **Acosta RJ**, et al. Validation of the Shoebox PureTest audiometry technology for remote data collection in clinical trials. *Scientific Reports*. doi: <https://doi.org/10.1038/s41598-026-49550-8>. Jun 2026
- **Acosta RJ**, et al. Optimizing Clinical Trials with Digital Composite Covariates: A Case Study in Auditory Sciences. *Journal of the American Academy of Audiology*. doi: <https://doi.org/10.3766/jaaa.250063>. May 2026
- Eloot S, ..., **Acosta RJ** et al. Predicting dialyzer fiber blocking is hard due to high inpatient variability and limited utility of thrombin generation markers. *Scientific Reports*. doi: <https://doi.org/10.1038/s41598-026-36507-0>. Jan 2026.
- Hall DA, Kwei K, **Acosta RJ**, et al. Validity of a Wearable Digital Insole for Assessing Gait ON and OFF in Parkinson's Disease. *Annals of Clinical and Translational Neurology*. doi: <https://doi.org/10.1002/acn3.70333>. Feb 2026.
- **Acosta RJ**, et al. Peripheral blood cells as prognostic markers in patients with advanced non-small cell lung cancer treated with cemiplimab as monotherapy or in combination with chemotherapy. *Cancer Treatment and Research Communications*. doi: <https://doi.org/10.1016/j.ctarc.2025.100959>. Jul 2025.
- Turner S, Chen C, **Acosta RJ**, et al. Methods for analytical validation of novel digital clinical measures: A simulation study. *medRxiv*. doi: <https://doi.org/10.1101/2024.11.29.24318211>. Dec 2024.
- Loewinger G, **Acosta RJ**, Mazumder R, Parmigiani G. Optimal ensemble construction for multistudy prediction with applications to mortality estimation. *Statistics in Medicine*. doi: [10.1002/sim.1006](https://doi.org/10.1002/sim.1006). Feb 2024.
- **Acosta RJ**, et al. All-cause excess mortality across 90 municipalities in Gujarat, India, during the Covid-19 pandemic (March 2020 - April 2021). *PLOS Global Public Health*. doi: [10.1371/journal.pgph.0000824](https://doi.org/10.1371/journal.pgph.0000824). Aug 2022.
- **Acosta RJ**, Irizarry RA. A Flexible Statistical Framework for Estimating Excess Mortality. *Epidemiology*. doi: [10.1097/EDE0000000000001445](https://doi.org/10.1097/EDE0000000000001445). May 2022.
- Kiang MV, **Acosta RJ**, et al. Sociodemographic and geographic disparities in excess fatal drug overdoses during the COVID-19 pandemic in California: A population-based study. *The Lancet Regional Health Americas*. doi: [10.1016/j.lana.2022.100237](https://doi.org/10.1016/j.lana.2022.100237). Mar 2022.

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- Nazrul I, Shkolnikov VM, **Acosta RJ**, et al. Excess deaths associated with Covid-19 pandemic in 2020: Age and sex desegregated time series analysis in 29 high income countries. *British Medical Journal*. doi: [10.1136/bmj.n1137](https://doi.org/10.1136/bmj.n1137). May 2021.
- **Acosta RJ**, Kishore N, Irizarry RA, Buckee C. Quantifying the dynamics of migration after Hurricane Maria in Puerto Rico. *PNAS*. doi: [10.1073/pnas.2001671117](https://doi.org/10.1073/pnas.2001671117). Dec 2020.

ABSTRACTS AND PRESENTATIONS

- Conventional Instruments Better Capture Progression of Parkinson's when Modeled Jointly: A Precursor to a Development Platform for Multimodal Composite Endpoints. *AD/PD 2026 Alzheimer's & Parkinson's Diseases Conference*. Copenhagen, Denmark. Poster presentation. Mar 2026.
- Gait Assessment with Digital Health Technologies in Parkinson's Disease with Levodopa Treatment. *Digital Health Technologies Summit at Regeneron*. New York, USA. Oral presentation. May 2025.
- Validation of Shoebox PureTest Audiometry in Regeneron Employees. *Digital Health Technologies Summit at Regeneron*. New York, USA. Poster presentation. May 2025.
- Gait Assessment with Digital Health Technologies in Parkinson's Disease with Levodopa Treatment. *SCOPE Summit*. Florida, USA. Oral presentation. Feb 2025.
- Gait Assessment with Digital Health Technologies in Parkinson's Disease with Levodopa Treatment. *International Congress of Parkinson's Disease and Movement Disorders*. Pennsylvania, USA. Poster presentation. Sep 2024.
- Prognostic Utility of Peripheral Myeloid Cells for Clinical Outcome in Patients with Non-Small Cell Lung Cancer (NSCLC) with Cemiplimab. *World Conference on Lung Cancer (WCLC)*. California, USA. Poster presentation. Sep 2024.
- Gait Assessment with Digital Health Technologies in Parkinson's Disease with Levodopa Treatment. *Digital Health Technologies Summit at Regeneron*. New York, USA. Poster presentation. May 2024.
- Developing Biomarkers for Auditory Science. *Digital Biomarkers in Clinical Trials Summit at Roche*. Basel, Switzerland. Oral presentation. May 2023.
- Developing Biomarkers for Auditory Science. *Digital Health Technologies Summit at Regeneron*. New York, USA. Oral presentation. May 2023.

PATENTS

- **Title:** Systems and Methods for Performing Gait and Speech Assessment Associated With a Disease State.
Patent Number: US20250292907A1
Date Issued: March 13, 2025
Patent Office: United States Patent and Trademark Office (USPTO)
Description: Machine learning models trained on wearable sensor and speech data can enable objective, continuous monitoring of neurological disease outside the clinic, replacing subjective, rater-dependent assessments with precise digital biomarkers. In this patent we present a framework for validating a low-cost wearable insole and a mobile speech-analysis platform against clinical gold-standard devices, using patients' own medication ON/OFF states as a built-in control to demonstrate that these simpler, portable tools can match gold-standard performance in detecting Parkinson's disease gait and speech impairment.
- **Title:** Systems and Methods for Reducing Sample Sizes.
Patent Number: US20240379195A1
Date Issued: May 9, 2024
Patent Office: United States Patent and Trademark Office (USPTO)

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Description: Machine learning and artificial intelligence models coupled with vast datasets can be used to make clinical trials shorter, cheaper, and smaller without loss of any scientific rigor. In this patent we present work that combines machine learning models and external datasets to augment clinical trials.